EAST POINT COLLEGE OF ENGINEERING & TECHNOLOGY
Department of CSE - (IoT & CSBT)
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Empowering visually impaired with Safe , Smart Navigation

Problem statement

Visually impaired individuals face significant challenges in navigating complex and dynamic environments safely. Traditional aids such as canes or human assistance often limit independence and efficiency. There is a pressing need for a smart, autonomous solution that ensures reliable mobility while reducing dependency. And addresses the problem by developing an Arduino-based obstacle avoiding robot using ultrasonic sensing, servo control, and motor drive automation, enabling safe navigation, enhancing accessibility, supporting broader applications in automation and surveillance.

Scope of the Project

The scope of this project includes designing, programming, and testing an Arduino-powered obstacle avoiding robot equipped with ultrasonic sensors, servo motors, and motor drive automation. It focuses on enabling visually impaired individuals to navigate safely and independently in dynamic environments. Beyond personal navigation, the project extends to applications in industrial automation, surveillance, and smart mobility systems. The scope emphasizes cost-effectiveness, adaptability, and efficiency, ensuring the robot can function reliably across diverse scenarios.

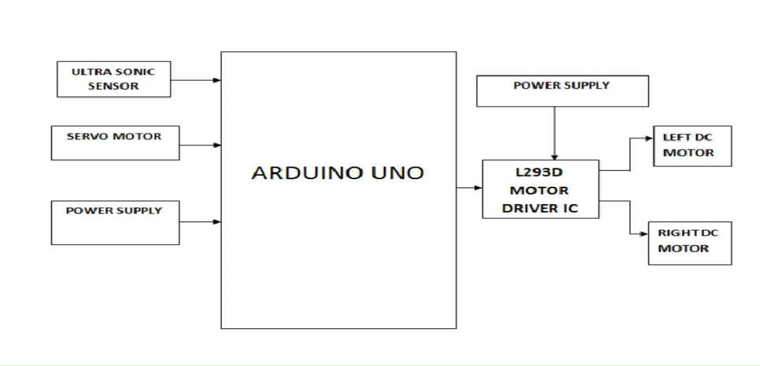
Applications

- **Assistive Navigation:** Provides safe, autonomous mobility for visually impaired individuals.
- **Surveillance Systems:** Enables patrolling and monitoring in restricted or hazardous areas.
- **Industrial Automation:** Supports material handling and movement in dynamic factory environments.
- **Smart Mobility:** Demonstrates autonomous navigation for future transport solutions.
- **Educational Tool:** Serves as a practical project for robotics and embedded systems learning.

Objectives

The objective of this project is to design and implement an Arduino-based obstacle avoiding robot that empowers visually impaired individuals with safe, autonomous navigation. By integrating ultrasonic sensors, servo motors, and motor drive automation, the system detects and avoids obstacles in real time. It aims to reduce reliance on human assistance, enhance accessibility, and promote independence. Additionally, the robot demonstrates applications in surveillance, industrial automation, and smart mobility, offering a versatile, cost-effective solution for dynamic environments

Architecture



Expected outcome

The expected result of this project is a fully functional Arduino-based obstacle avoiding robot capable of autonomously navigating dynamic environments. Using ultrasonic sensors, servo control, and motor drive automation, the robot will detect and avoid obstacles in real time. It will provide safe, reliable mobility for visually impaired individuals, reducing dependence on human assistance.

Abstract

This project presents an Arduino-based obstacle avoiding robot designed to assist visually impaired individuals with safe navigation. Using ultrasonic sensors, servo control, and motor drive automation, the robot autonomously detects and avoids obstacles in dynamic environments. It ensures reliable mobility, reduces dependence on human assistance, and enhances accessibility. Beyond personal navigation, the system demonstrates applications in surveillance, industrial automation, and smart mobility, offering a cost-effective, efficient solution for autonomous movement and improved safety in complex surroundings.

Tools Used:

->**Hardware Tools**
- Arduino Uno
- Ultrasonic Sensor (HC-SR04 / Ping)
- Servo Motor (SG90 / MG995)
- Motor Driver Shield (L298N)
- DC Motors with Wheels
- Power Supply (9V Battery / Li-Ion Pack)
- Chassis and Mounting Accessories

->**Software Tools**
- Arduino IDE
- Ping Library
- Servo Library
- Motor Drive Shield Library

Team Members

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